

SHIVAM KUMAR

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Education

National Institute of Technology, Rourkela

Bachelor of Technology in Industrial Design Engineering (CGPA: 7.60/10)

Aug 2023 – Present

Rourkela, Odisha

Mahatma Gandhi High School, Jhajha (BSEB)

AISCCE – Science (PCM) – 82.8%

April 2023

Jhajha, Bihar

Mahatma Gandhi High School, Jhajha (BSEB)

AISSE – 89.8%

April 2021

Jhajha, Bihar

Relevant Coursework

- Data Structures and Algorithms
- Database Management Systems
- Web Development
- Object Oriented Programming
- SQL
- Statistics & Probability

Projects

Social Network Friend Suggestion System

Oct 2025

C++, OOPS

[GitHub](#)

- Engineered a scalable **graph-based social network engine** using **adjacency list representation**, modeling **100+ users** and **500+ bidirectional edges**, reducing memory usage by **40%** compared to matrix representation.
- Developed a **Breadth-First Search (BFS)** powered recommendation system to compute **2nd-degree connections** with optimized traversal complexity of **$O(V + E)$** , enabling real-time query execution under **50ms**.
- Devised a **priority-queue-based ranking strategy** combining mutual-friend count and node degree to return the **top 5 most relevant friend suggestions per user**.
- Incorporated **file-based persistence** to store and reload user data, enabling **100% data recovery** across **multiple program executions**.

Catch'n Match – Full Stack Interactive Web Game

Aug 2025

React.js, Node.js, Express.js, MongoDB, JWT Auth, Bootstrap, Framer Motion

[GitHub](#)

- Built a **full-stack web game** featuring **3+ gameplay levels**, implementing shape recognition logic and **time-bound real-time object interaction**.
- Designed **JWT-based authentication and authorization** to support **secure login sessions** and personalized gameplay for **registered users**.
- Built and integrated **RESTful APIs** using **Node.js** and **Express.js**, achieving average API response time below **120ms**.

Achievements & Certifications

- **Certified MERN Stack Developer – Apna College {Certificate}** June 2025
Completed **120+** hours of hands-on full-stack training covering **React.js, Node.js, Express.js, MongoDB, RESTful APIs**, authentication, and deployment.
- **LeetCode 100 Days Coding Challenge – LeetCode {Badge}** Jan 2025
Solved **100+ Data Structures and Algorithms problems** in C++ and JavaScript, ranking in the top 15% among global participants.

Technical Skills

- **Languages:** C++, Python, JavaScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** Node.js, React.js, Express.js, Tailwind CSS, Bootstrap, MongoDB
- **Tools & Platforms:** Git, GitHub, VS Code

Extracurricular Activities

All India Inter NIT Sports Fest

Oct 2025

Core Team Member

- Led the coordination of **10+ national-level sporting events**, managing logistics, participant onboarding, and on-site operational execution.

Management Head – Intra Hall Events, VS Hall, NIT Rourkela

Jan 2024

Management Head

- Supervised intra-hall events by managing **15+ organizers and volunteers**, ensuring operational efficiency and timely completion of activities.